Daphne Qin

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Education

Rice University Aug 2023 – May 2027

Bachelor of Science in Computer Science, GPA 3.8/4.0 *Houston, TX*

- **Relevant Coursework**: Game Design, Data Structures and Algorithms, Concurrent Programming, Object-Oriented Design, Computer Engineering, Computer Architecture, Neuroengineering
- **Awards & Honors**: Tau Beta Pi (2026), Data Mentoring 2nd Place (2025), President's Honor Roll (2023/2024), Data Showcase 1st Place (2023), Certificate for Excellence in Technology (2023), Franklin Scholarship (2023-2024)

Professional Experience

Active Safety Simulation Developer May 2026 – Aug 2026

General Motors *Warren, MI*

- Pioneered a pipeline that automatically generates isolated subset models of any car's active safety controller in under 10 seconds.
- Extended in-house simulation scripting language by implementing parsing in C for test groups and boolean logic, thus enhancing regression testing workflows aiding over 60 lead calibrators and software developers in active safety teams.
- Shipped software releases ahead of schedule by 3 weeks using Agile-style methodology.

Computational Biology Researcher Jun 2024 – May 2026

Baylor College of Medicine *Houston, TX*

- Co-author and contributor to the latest paper by the ENCODE consortium, to be published by ***Nature*** in 2026.
- Designed and implemented a multi-stage data processing pipeline to analyze millions of lines of uncharacterized data across over 800 proteins using Python and Bash script.
- Optimized data analysis runtime by 50% by introducing SLURM-based parallel job scheduling and concurrency optimization on an HPC cluster.
- Improved workflow for 10 professors and researchers, cutting cycle time by 20%, by developing interactive protein-to-genome relationship visualization platform using Flask.
- Presented technical findings to audiences of 100+ researchers and interdisciplinary stakeholders.

Machine Learning Research Assistant Jan 2024 – May 2026

Baylor College of Medicine *Houston, TX*

- Improved brain structure classification efficiency by 100% by developing a semi-supervised machine learning algorithm using Python scikit-learn.
- Enhanced classification accuracy by 15% by designing a novel estimation and flood-filling algorithm.

Peer Tutor Aug 2024 – May 2025

Rice University *Houston, TX*

- Led a team of 50 TAs to organize a campus-wide interactive simulation bringing the A-star algorithm to life.
- Mentored over 300 students on concepts including graph search algorithms and predictive model fitting.

Software Engineer Jan 2024 – Apr 2024

American Medical Women's Association *Houston, TX*

- Optimized physician lookup times by 20% by implementing a server-side database caching system.
- Contributed to a platform later featured on Fox 26 Houston and ABC 13, reaching college students nationwide.

Projects

AI-Driven Narrative Game | *Godot, Google AI Studio* | Itch.io, GitHub Nov 2025 – May 2026

- Designed and implemented a modular game system integrating AI-generated character interactions with player-driven narrative state using Godot and custom Google AI integration.
- Built persistent dialogue memory, state tracking, and rule-based safety constraints to ensure deterministic, coherent character behavior across gameplay sessions.
- Engineered a low-latency signaling pipeline that parses AI-generated keywords to trigger in-game animations, decoupling game logic from AI services and improving responsiveness.

Technical Skills

Presentation: Public Speaking, Conflict Management (Doerr Institute Workshop)
Languages: Proficient in Python, Java, C, C++; Familiar with JavaScript, TypeScript, HTML/CSS, Shell, SQLite
Frameworks: REST APIs, Flask, JUnit, Cypress
Developer Tools: Git, Google AI Studio, VS Code, Visual Studio, PyCharm, IntelliJ, Godot
Libraries: Pandas, NumPy, MatPlotLib, scikit-learn